EV MARKET ANALYSIS USING POWERBI

Project Title : EV Market Analysis

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INTRODUCTION

Overview

- Electric Vehicles (EVs) are transforming the transportation industry.
- Growing demand due to environmental concerns and government incentives.
- Data-driven analysis helps understand market trends and consumer behavior.

Purpose

- Analyze EV market data using Power BI.
- Generate insights for decision-making and future market predictions.

Problem Statement

Challenges in EV Market Analysis

- Large volumes of market data from multiple sources.
- Difficulty identifying sales trends and market growth patterns.
- Need for visual dashboards for quick decision-making.

Solution

- Use Power BI to clean, analyse, and visualize EV market data.

Project objectives

- Analyze EV sales trends.
- Identify top-performing EV manufacturers and models.
- Study regional adoption patterns.
- Evaluate market growth over time.
- Create interactive dashboards using Power BI.

Dataset Description

- Data Source : ev_market_2026.csv
- Web Site : www.Kaggle.com
- Table contains 2000 records and 22 key attributes.

  (2000 rows and 22 columns)

Column Definitions

Column	Type	Range	Description
brand	categorical	20 brands	Tesla, BYD, Rivian, Lucid, NIO, Hyundai, Kia, BMW, Mercedes, Audi, Volkswagen, Ford, GM/Chevrolet, Toyota, Polestar, Fisker, Porsche, Volvo, Xiaomi, Honda
model	categorical	~5/brand	Tesla Model 3/Y/S/X/Cybertruck, BYD Seagull/Qin/Atto 3, Rivian R1T/R1S, etc.
year	integer	2020-2026	Model year (weighted toward recent years)
variant	categorical	5 types	Base, Standard, Long Range, Performance, Premium
price_usd	float	$25K-$200K	TARGET VARIABLE - Realistic pricing based on brand, model, variant, year
battery_capacity_kwh	float	40-120	Battery size (correlated with range)
range_miles	float	150-450	EPA/WLTP range (strongly correlated with battery)
charging_speed_kw	float	50-350	DC fast charging capability
acceleration_0_60_mph	float	2.0-8.0	0-60 seconds (performance variants faster)
top_speed_mph	float	100-200	Maximum speed capability
horsepower	integer	150-1000	Total system power
torque_nm	integer	200-1200	Maximum torque in Newton-meters
drive_type	categorical	3 types	AWD, FWD, RWD
seating_capacity	integer	2-7	Number of passenger seats
body_type	categorical	6 types	Sedan, SUV, Truck, Hatchback, Coupe, Van

Column Definitions

cargo_volume_cubic_ft	**float**	**15-80**	**Trunk/cargo space**
weight_kg	float	1500-3000	Vehicle curb weight
safety_rating	integer	3-5	NCAP safety stars (most EVs: 4-5)
autopilot_level	integer	0-3	SAE automation level (Tesla/Mercedes: 2-3)
country_of_origin	categorical	6	US, China, Germany, South Korea, Sweden, Japan
market_segment	categorical	4	Budget, Mid-range, Premium, Luxury
annual_sales_units	integer	500-500K	Annual production volume
customer_rating	float	3.0-5.0	Customer satisfaction (correlated with range)
warranty_years	integer	3-8	Warranty period

Tools And Technologies used

Software

- Microsoft Power BI
- Microsoft Excel
- Power Query

Techniques

- Data Cleaning
- Data Transformation
- Data Visualization
- Dashboard Development

Methodology

1. Data Collection
2. Data Cleaning
3. Data Transformation
4. Data Modeling
5. Dashboard Creation
6. Analysis and Reporting

1. **Flow Diagram:**
   Data Source → Power Query → Data Model → Visualization → Insights

Data Preparation

Steps Performed

- Removed duplicates
- Handled missing values
- Standardized formats
- Created calculated columns and measures

Benefits

- Improved data quality
- Accurate analysis results

DAX Measures

- Average_Battery_Capacity
- Average_Customer_Rating
- Average_Horse_Power
- Average_Miles_Range
- Average_Vehicle_Price
- Average_Safety_Rating
- High_Safety_Vehicles
- Market_Share%
- Revenue_Per_Vehicle
- Total_Revenue
- Total_EV_Sales
- Unique_Brands
- Uniqe_Models
- YoY Sales_Growth%

Calculated Columns

- Battery_Capacity_Cagtegory
- Charging_Speed_Status
- Customer_Rating_Category
- Day
- Month_Name
- Model_Year
- Efficiency_Metric
- Model_Year
- Range_Category
- Safety_Category
- Sales_Performance
- Total_Price_Annum

Dashboard Overview

Dashboard Components

- Total EV Sales
- Unique brands
- Unique Models
- Total Revenue
- Market Share%
- YoY Sales Growth%
- Average Vehicle Price
- Average Range Miles
- Average Customer Rating

Possible Slicers added

- Brand
- Model
- Varient
- Year
- Battery_Capacity_Category
- Market_Segment

Navigations Inserted

Bookmark Navigation

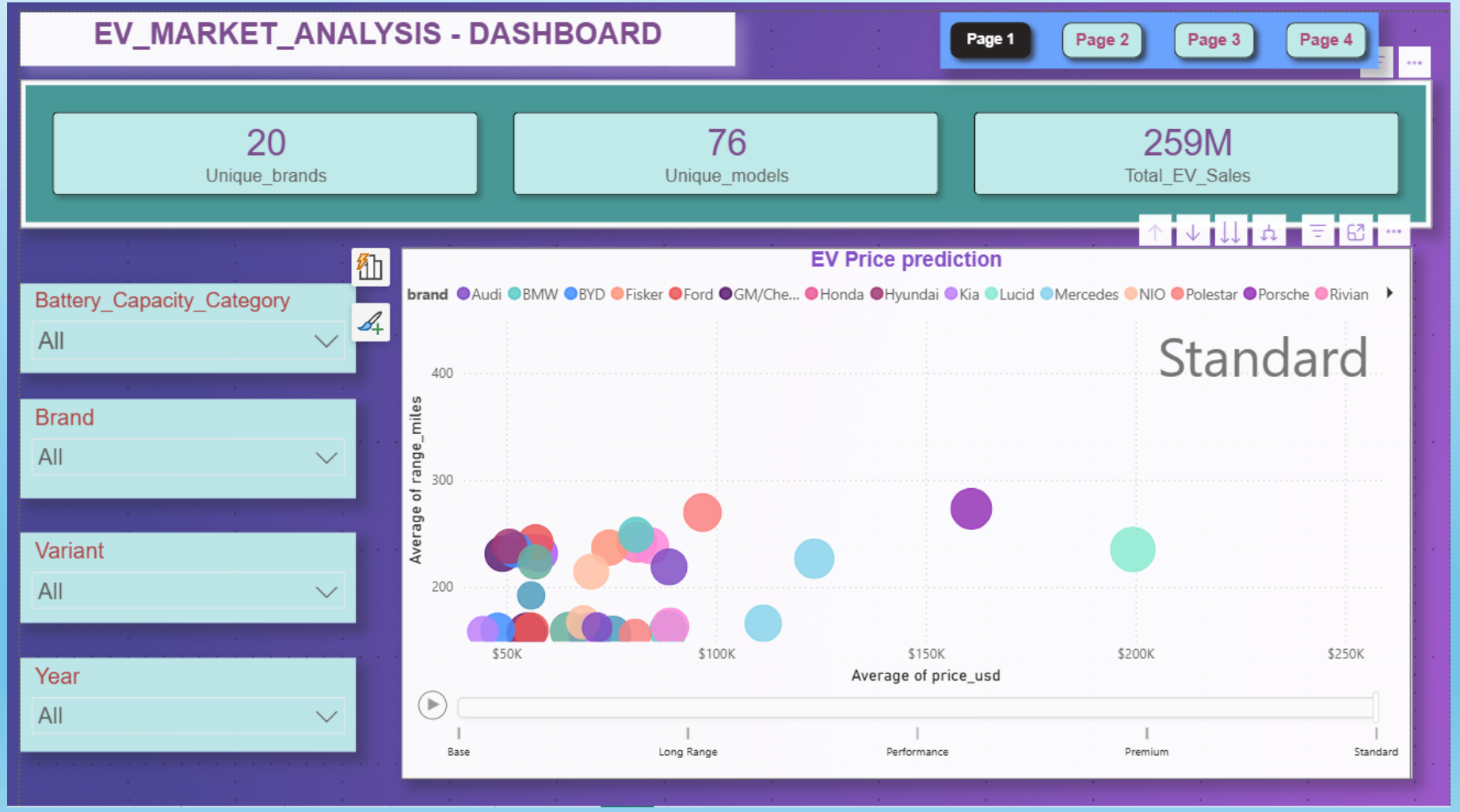
1. Efficiency Metric (Matrix inserted in page 2)
2. Value Analysis Charts (Page 2)
3. Brand Comparison Matrix (Page 3)
4. Brand Comparison Charts (Page 3)

Page Navigation

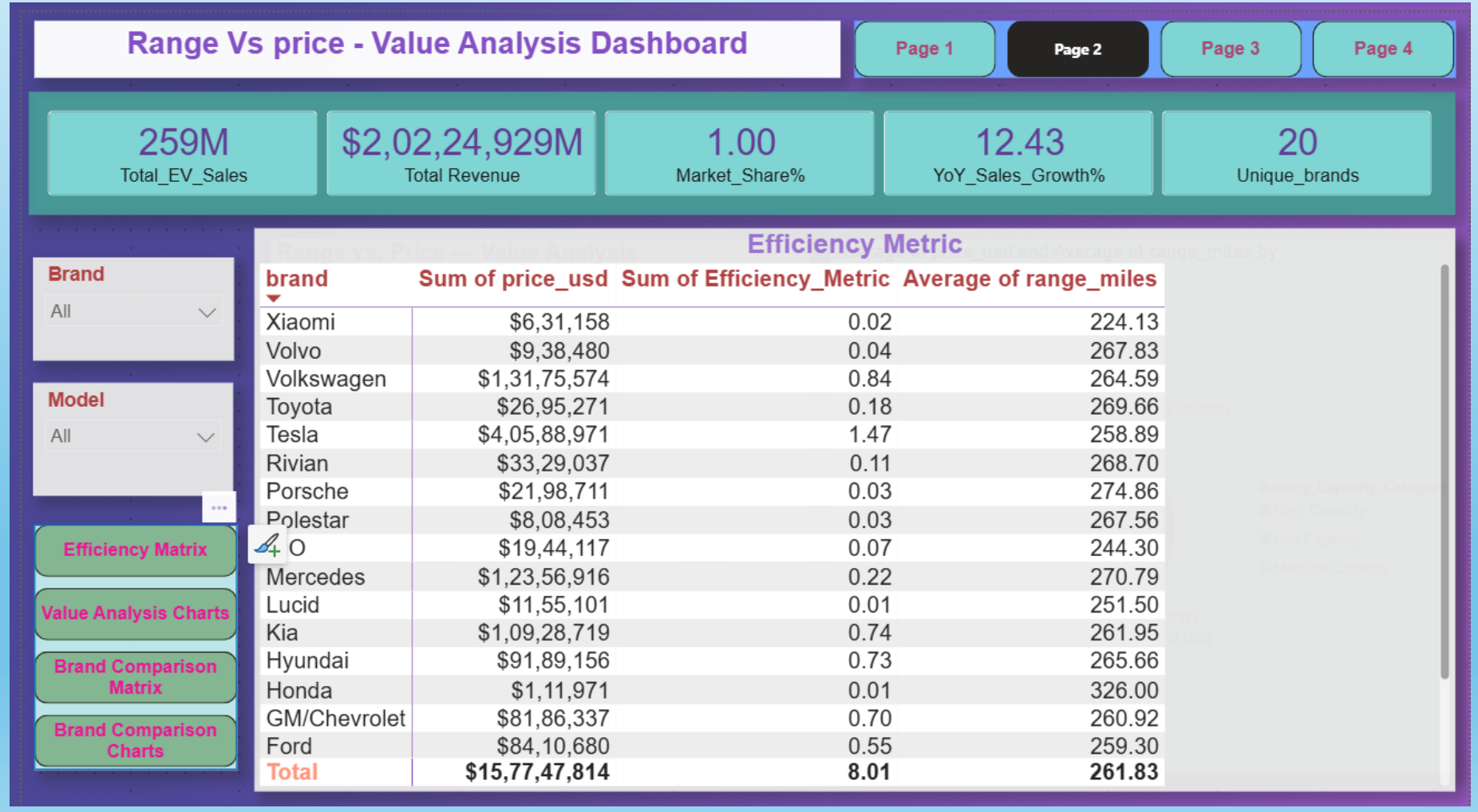
1. Inserted Page Navigations to all pages to navigate through pages

DASHBOARD SCEENSHOTS

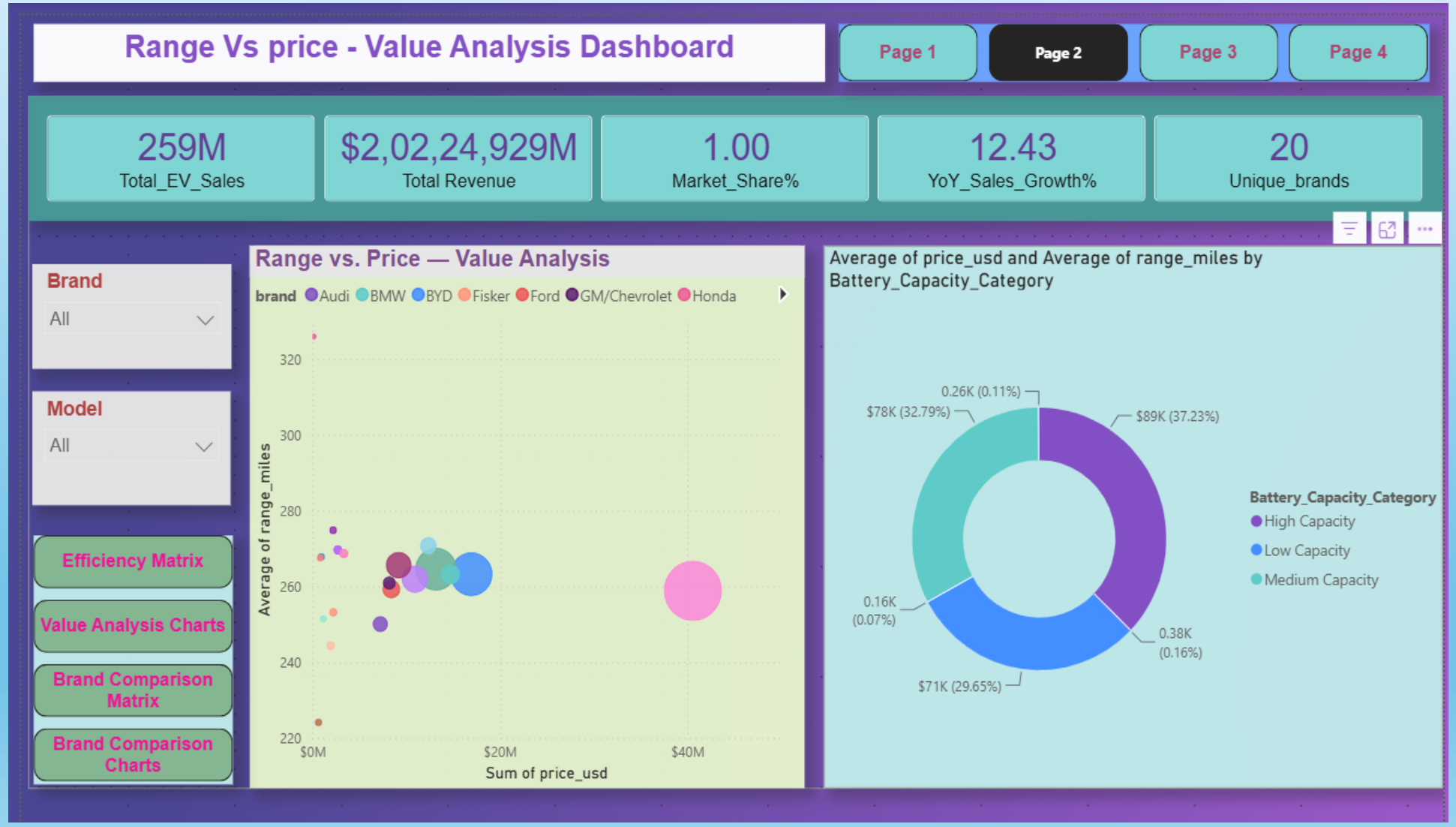
Page 1- EV PRICE PREDICTION (Scatter Chart)



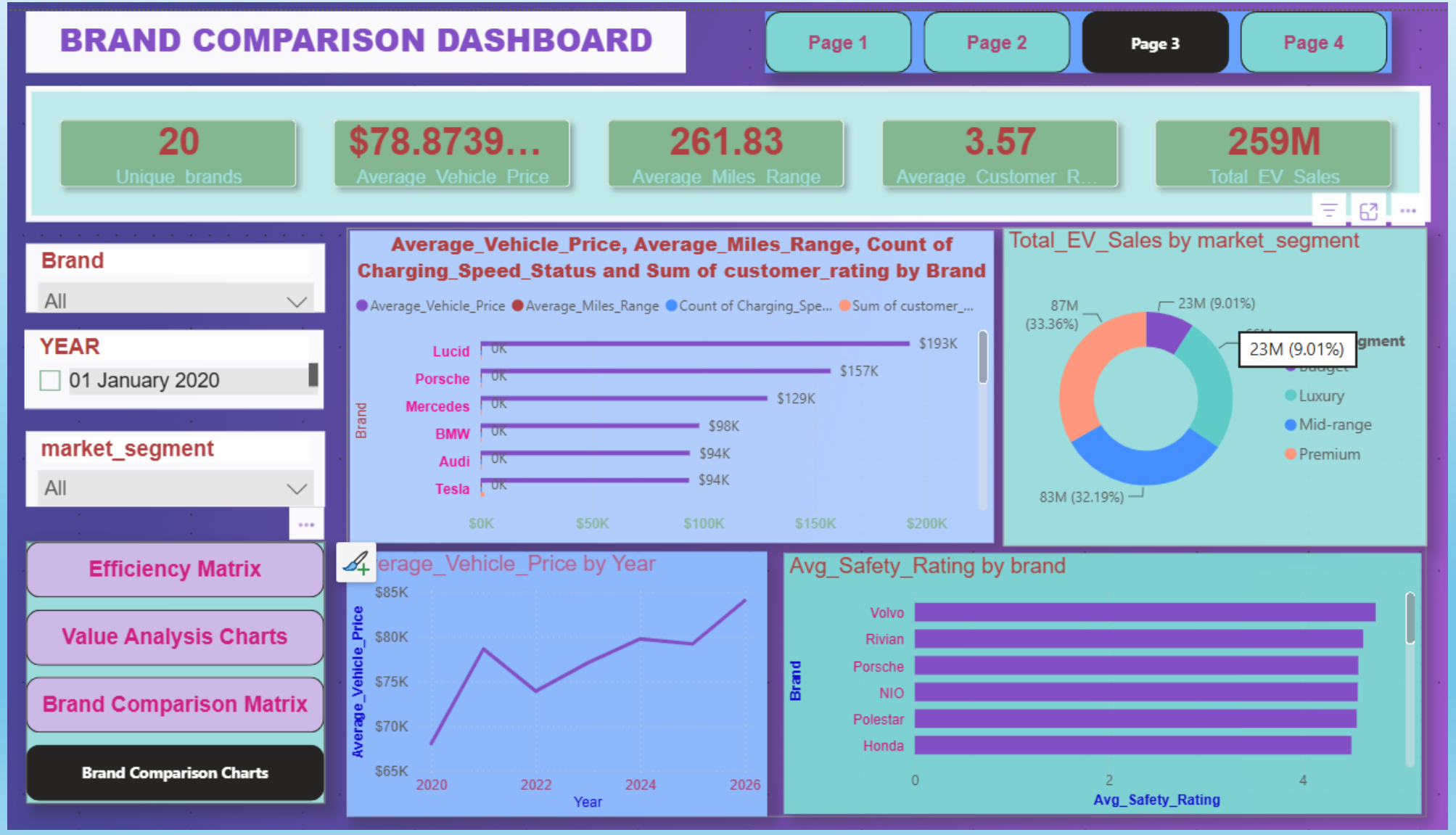
Page 2(1) Efficiency Matrix



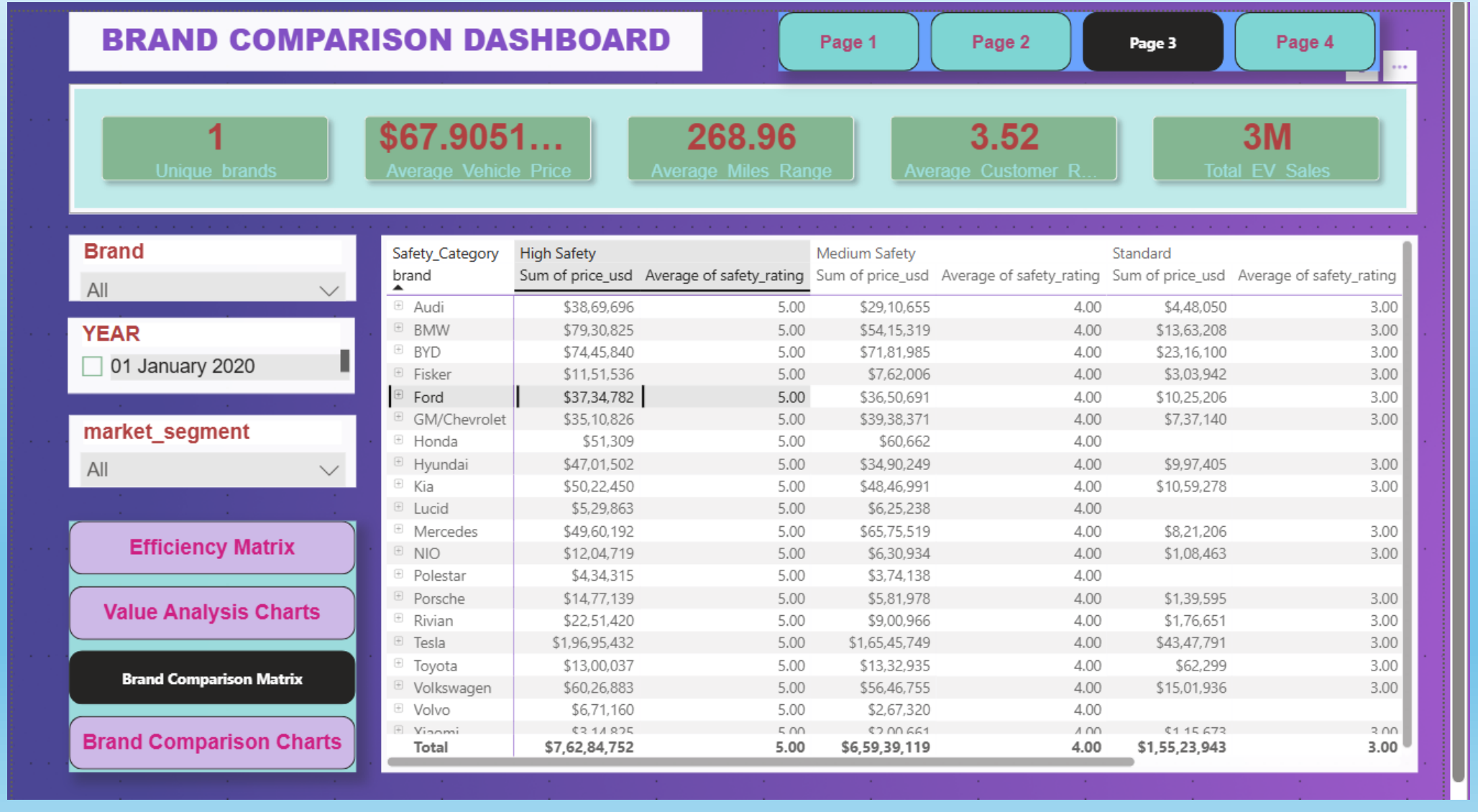
Page 2 (2) Value Analysis Charts



Page 3(1) Brand Comparison Charts

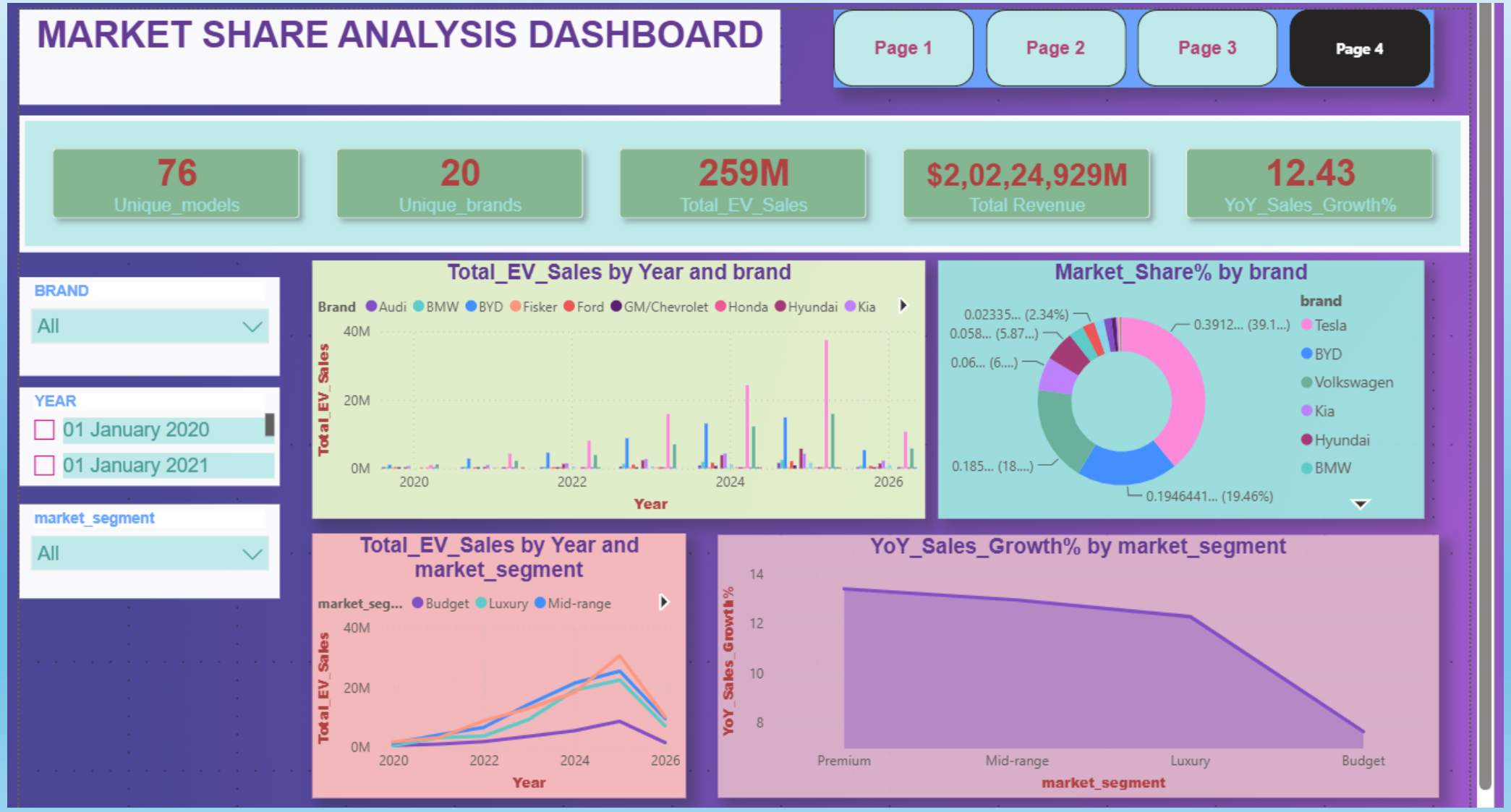


Page 3(2) Brand Comparison Matrix



Safety_Category	High Safety		Medium Safety		Standard	
brand	Sum of price_usd	Average of safety_rating	Sum of price_usd	Average of safety_rating	Sum of price_usd	Average of safety_rating
Audi	$38,69,696	5.00	$29,10,655	4.00	$4,48,050	3.00
BMW	$79,30,825	5.00	$54,15,319	4.00	$13,63,208	3.00
BYD	$74,45,840	5.00	$71,81,985	4.00	$23,16,100	3.00
Fisker	$11,51,536	5.00	$7,62,006	4.00	$3,03,942	3.00
Ford	$37,34,782	5.00	$36,50,691	4.00	$10,25,206	3.00
GM/Chevrolet	$35,10,826	5.00	$39,38,371	4.00	$7,37,140	3.00
Honda	$51,309	5.00	$60,662	4.00		
Hyundai	$47,01,502	5.00	$34,90,249	4.00	$9,97,405	3.00
Kia	$50,22,450	5.00	$48,46,991	4.00	$10,59,278	3.00
Lucid	$5,29,863	5.00	$6,25,238	4.00		
Mercedes	$49,60,192	5.00	$65,75,519	4.00	$8,21,206	3.00
NIO	$12,04,719	5.00	$6,30,934	4.00	$1,08,463	3.00
Polestar	$4,34,315	5.00	$3,74,138	4.00		
Porsche	$14,77,139	5.00	$5,81,978	4.00	$1,39,595	3.00
Rivian	$22,51,420	5.00	$9,00,966	4.00	$1,76,651	3.00
Tesla	$1,96,95,432	5.00	$1,65,45,749	4.00	$43,47,791	3.00
Toyota	$13,00,037	5.00	$13,32,935	4.00	$62,299	3.00
Volkswagen	$60,26,883	5.00	$56,46,755	4.00	$15,01,936	3.00
Volvo	$6,71,160	5.00	$2,67,320	4.00		
Xiaomi	$3,14,825	5.00	$2,00,661	4.00	$1,15,673	3.00
Total	**$7,62,84,752**	**5.00**	**$6,59,39,119**	**4.00**	**$1,55,23,943**	**3.00**

Page 4 Market Share Analysis



Key Insights

- EV adoption is steadily increasing.
- Certain Models and Brands lead the market.

Benefits of Power BI

- Interactive dashboards
- Real-time data analysis
- Easy report sharing
- Advanced visualizations
- Better business decision-making

Challenges and Limitations

- **Challenges**
- Data consistency issues
- Rapidly changing market conditions

- **Limitations**
- Analysis depends on available historical data.

Future Scope

- Predictive analysis using Machine Learning.
- Integration with real-time EV market data.
- Customer behaviour analysis.
- Advanced forecasting dashboards.

Conclusion

- Successfully analyzed EV market data using Power BI.
- Identified trends, growth opportunities, and market leaders.
- Demonstrated how data visualization supports strategic decisions.

References

- Government EV reports
- Industry market reports
- Power BI Documentation
- EV market datasets

THANK YOU